



ADVANCED TRACK — WEEK 06-07 COMBINED ASSIGNMENT

Threat Intelligence Automation + Infrastructure as Code Security

Track: AI, Automation & Security Engineering

Combined Assignment: Week 06-07

Threat Intel Automation + Infrastructure Security Engineering

Company	THE ARZENS	Program	Engineering Internship — Advanced Track
Track	AI, Automation & Security Engineering	Deadline	Thursday, 12 September 2026, 11:59 PM (submitted via Google Classroom)
Total Points	30 (5 points per task, 6 tasks total)	Submission Format	One combined ZIP: 6 tasks total (3 from Week 06 + 3 from Week 07)
Submission Mode	Individual — Google Classroom assignment upload only		

Overview

This combined assignment covers enterprise security automation: threat intelligence enrichment pipelines (Week 06) and secure Infrastructure as Code with Terraform & Ansible (Week 07). You will build production-grade TI platforms that query multiple sources, manage IOC lifecycles, and secure cloud infrastructure through code. This is real-world security engineering — no simulations, actual implementation.

Part A: Week 06 Content — Threat Intelligence Automation (Tasks 1–3)

Task 1: Theoretical — TI Platform Architecture Design

5 Points

Design threat intelligence enrichment architecture for enterprise SOC.



Requirements:

- Length: 800–1000 words
- Components: Data sources (internal/external), enrichment engine, IOC database, workflow integration
- Address: Rate limit handling, IOC expiration, false positive prevention
- Include architecture diagram with data flow
- Reference: MISP Project (misp-project.org)

Deliverable: PDF document with diagram

Task 2: Technical — TI Enrichment Engine

5 Points

Build Python tool that enriches indicators using multiple TI sources.

External Resources:

- [EXT-1] VirusTotal API Key — virustotal.com (free tier)
- [EXT-2] AbuseIPDB API Key — abuseipdb.com (free tier)
- [EXT-3] AlienVault OTX — otx.alienvault.com (free)

Implementation:

- Input: Single indicator (--indicator) or batch (--input-file)
- Query 3 sources: VirusTotal, AbuseIPDB, OTX
- Rate limit management: Respect API limits, implement caching
- Risk scoring algorithm: 0-100 based on source responses
- Output: Console table, JSON, CSV formats
- Error handling: API failures, timeouts, invalid responses

Deliverables:

- ti_enricher.py
 - config.yaml (API keys template)
 - cache.json (query cache)
 - sample_indicators.csv
 - enrichment_results.csv
-

Task 3: Technical — IOC Manager & Automation

5 Points

Build system to manage enriched IOCs with lifecycle management.

Implementation:

- IOC database: JSON storage with metadata (first_seen, last_seen, expiration, confidence)
- Lifecycle: Add, update, expire IOCs automatically
- Confidence scoring: Rule-based (source diversity, score stability, recency)



- Workflow integration: SIEM blocklist export, analyst reports, alerting
- Automation: Scheduled updates, expiration checks, health monitoring

Commands:

- --add-file: Import new IOCs
- --update-all: Re-enrich all IOCs
- --expire-check: Remove stale IOCs
- --export-blocklist: Generate SIEM import

Deliverables:

- ioc_manager.py
- ioc_database.json
- ioc_config.yaml
- blocklist.txt (sample export)
- weekly_report.html (sample report)

Part B: Week 07 Content — Infrastructure as Code Security (Tasks 4–6)

Task 4: Theoretical — IaC Security Architecture

5 Points

Design secure Infrastructure as Code architecture for cloud deployment.

Requirements:

- Length: 800–1000 words
- Tools: Terraform (provisioning) + Ansible (configuration)
- Security controls: State file encryption, secret management, policy enforcement
- Address: Drift detection, rollback procedures, compliance validation
- Include: Terraform module structure diagram
- Reference: HashiCorp Security Best Practices

Deliverable: PDF document with architecture diagram

Task 5: Technical — Terraform Secure Infrastructure

5 Points

Build Terraform configuration for secure cloud infrastructure.

External Resources:

- [EXT-4] Terraform Cloud — terraform.io (free tier)
- [EXT-5] AWS Free Tier — aws.amazon.com/free (optional, can use local)

Implementation:



- Provider: AWS (or localstack for testing)
- Resources: VPC, security groups, EC2 instance, S3 bucket
- Security hardening:
 - Security group: Restrict ports (22, 80, 443 only)
 - S3 bucket: Private, encryption enabled, versioning
 - EC2: Key pair auth, no root login, updated AMI
- State management: Remote backend with encryption
- Variables: Externalized configuration (variables.tf)
- Outputs: Resource IDs, endpoints

Files:

- main.tf (core infrastructure)
- variables.tf (configurable inputs)
- outputs.tf (resource outputs)
- terraform.tfvars (sensitive values, never commit)

Deliverables:

- main.tf, variables.tf, outputs.tf
 - terraform.tfvars.example
 - README.md (deployment instructions)
 - terraform_plan.txt (sample plan output)
-

Task 6: Technical — Ansible Hardening & Compliance

5 Points

Build Ansible playbooks for secure configuration management.

Implementation:

- Inventory: hosts.ini with server groups
- Playbooks:
 - site.yml: Main playbook importing all roles
 - security.yml: Security-specific tasks
- Roles:
 - common: System updates, timezone, logging
 - security: Firewall (ufw/iptables), fail2ban, auditd
 - compliance: CIS benchmarks checks, reporting
- Tasks:
 - Package updates (unattended-upgrades)
 - SSH hardening (disable root, key-only auth)
 - Firewall configuration (deny all, allow specific)
 - Audit logging (auditd rules for compliance)



- Compliance check: Generate report of passed/failed checks

Idempotency: Playbooks can run multiple times safely

Deliverables:

- site.yml (main playbook)
 - hosts.ini (inventory)
 - roles/ directory (common, security, compliance)
 - ansible.cfg (configuration)
 - compliance_report.txt (sample output)
 - ANSIBLE_README.md (execution guide)
-

External Resources Summary

[EXT-1] VirusTotal API:

- URL: [virustotal.com](https://www.virustotal.com)
- Cost: FREE (4 requests/minute)
- Signup: Required

[EXT-2] AbuseIPDB API:

- URL: abuseipdb.com
- Cost: FREE (1,000 requests/day)
- Signup: Required

[EXT-3] AlienVault OTX:

- URL: otx.alienvault.com
- Cost: FREE (unlimited with account)
- Signup: Optional

[EXT-4] Terraform:

- URL: terraform.io
- Cost: FREE (open source)
- Cloud: Free tier available

[EXT-5] AWS Free Tier:

- URL: aws.amazon.com/free
 - Cost: FREE (12 months)
 - Alternative: Use localstack for local testing
-

Submission Checklist — Week 06-07 Combined

- Task 1: TI Architecture Design (PDF)



THE ARZENS (TAS)

Engineering Internship Program — AI, Automation & Security Engineering Track

- Task 2: TI Enrichment Engine (Python + configs)
- Task 3: IOC Manager (Python + database)
- Task 4: IaC Security Architecture (PDF)
- Task 5: Terraform Configuration (.tf files)
- Task 6: Ansible Playbooks (.yaml files + roles)
- Git repository with all files
- AI Assistance Note (if applicable)

Deadline: Thursday 12 September 2026, 11:59 PM

THE ARZENS — Internship Program | Advanced Track | AI, Automation & Security Engineering